Stochastic Kriging for Large Data Sets

Liang DING

March 22, 2018

Hong Kong University of Science and Technology

Table of Contents

1. Gaussian Process (GP)
2. Markovian Structure and Sparsity
3. Sturm-Liouville (S-L) Problem
4. GP with Derivative Information
5. Weak Derivative of GP
6. Numerical Experiments
7. Conclusions

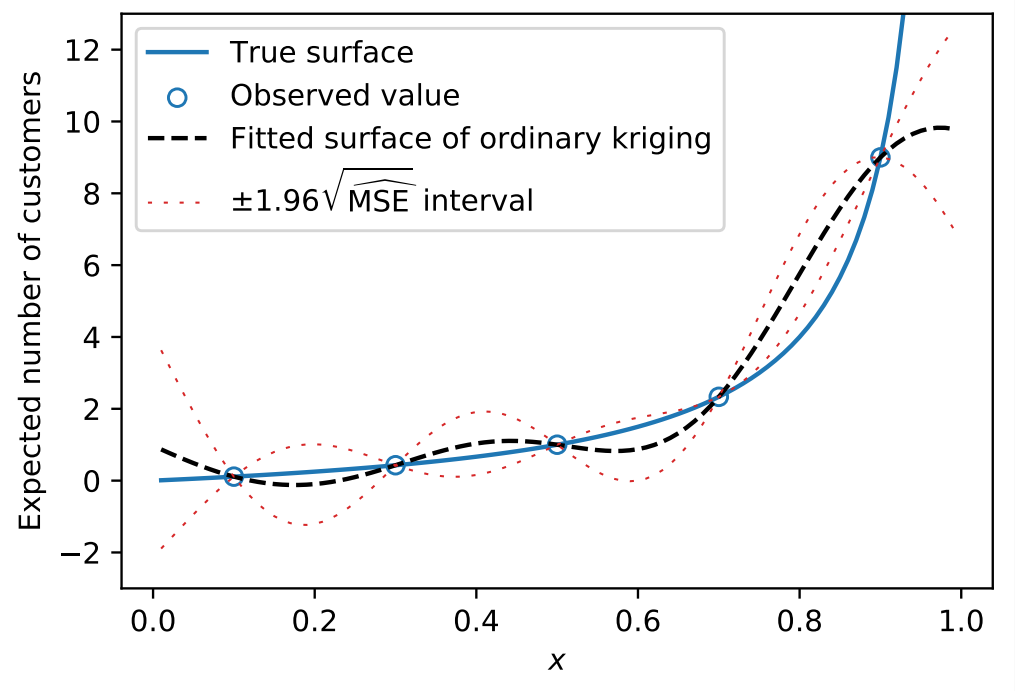
Gaussian Process (GP)

Gaussian Process

- Generalization of multivariate normal distribution
  - $(\mathrm{M}(\boldsymbol{x}_1), \ldots, \mathrm{M}(\boldsymbol{x}_n))$ is multivariate normal
- Completely characterized by covariance function $k(\boldsymbol{x}, \boldsymbol{y})$
- Unknown function surface is viewed as a GP realization

$$\mathrm{Z}(\boldsymbol{x}) = \beta + \mathrm{M}(\boldsymbol{x}), \quad \boldsymbol{x} \in \mathcal{X} \subseteq \mathbb{R}^d$$

- Leverage *spatial correlation* for prediction



Applications

- Spatial statistics
  - kriging
- Design and analysis of computer experiments
  - efficient global optimization
- Stochastic simulation
  - stochastic kriging
  - simulation optimization
- Machine learning
  - Gaussian process regression
  - Bayesian optimization

The Big n Problem

- Function surface is observed at $\{\boldsymbol{x}_1, \ldots, \boldsymbol{x}_n\}$ with noise

$$z_i = \beta + \mathrm{M}(\boldsymbol{x}_i) + \varepsilon(\boldsymbol{x}_i)$$

- The best linear unbiased predictor (BLUP) of $\mathrm{Z}(\boldsymbol{x}_0)$ is

$$\widehat{\mathrm{Z}}(\boldsymbol{x}_0) = \beta + \boldsymbol{\Sigma}_{\mathrm{M}}(\boldsymbol{x}_0, \cdot)[\boldsymbol{\Sigma}_{\mathrm{M}} + \boldsymbol{\Sigma}_{\varepsilon}]^{-1}[\overline{\boldsymbol{z}} - \beta \mathbf{1}_n]$$

- Slow: $[\boldsymbol{\Sigma}_{\mathrm{M}} + \boldsymbol{\Sigma}_{\varepsilon}] \in \mathbb{R}^{n \times n}$ and inverting it takes $\mathcal{O}(n^3)$ time
- Numerically instable: $[\boldsymbol{\Sigma}_{\mathrm{M}} + \boldsymbol{\Sigma}_{\varepsilon}]$ is often nearly singular

Approximation Schemes

- Well developed in spatial statistics and machine learning
  - Banerjee, Carlin, and Gelfand (2015)
  - Rasmussen and Williams (2006)
- Reduced-rank approximations
  - Emphasize large-scale dependences, but fail to capture small-scale dependences

  Sparse approximations (e.g., covariance tapering)
  - Emphasize small-scale dependences, but fail to capture large-scale dependences

Markovian Structure and Sparsity

Gaussian Markov Random Field (GMRF)

- Specify $\boldsymbol{\Sigma}_{\mathrm{M}}^{-1}$, instead of $k(\cdot, \cdot)$
- Use graph to describe Markovian structure
  - Given all its neighbors, node i is *conditionally independent* of its non-neighbors
  - E.g., $\mathrm{M}(\boldsymbol{x}_2) \perp (\mathrm{M}(\boldsymbol{x}_0), \mathrm{M}(\boldsymbol{x}_4))$, given $(\mathrm{M}(\boldsymbol{x}_1), \mathrm{M}(\boldsymbol{x}_3))$

- Sparsity: $\boldsymbol{\Sigma}_{\mathrm{M}}^{-1}(i, j) \neq 0 \iff i$ and j are neighbors
- The sparsity can reduce necessary computation to $\mathcal{O}(n^2)$

Disadvantages of GMRF

- Hard to specify desired correlation behavior
- Cannot predict locations "off the grid"

$$\widehat{Z}(\boldsymbol{x}_0) = \beta + \underbrace{\boldsymbol{\Sigma}_{\mathrm{M}}(\boldsymbol{x}_0, \cdot)}_{\text{unknown}}[\boldsymbol{\Sigma}_{\mathrm{M}} + \boldsymbol{\Sigma}_{\varepsilon}]^{-1}[\overline{\boldsymbol{z}} - \beta \mathbf{1}_n]$$

- Continuous design space must be discretized first, which may result in $N \gg n$ grid points
  - Computing predictor requires $\mathcal{O}(N^2)$
  - $\mathcal{O}(N^2)$ v.s. $\mathcal{O}(n^3)$?

The Best of Two Worlds?

- We will construct a class of covariance functions for which:
  1. $\boldsymbol{\Sigma}_{\mathrm{M}}$ can be inverted analytically
  2. $\boldsymbol{\Sigma}_{\mathrm{M}}^{-1}$ is sparse
- Explicit link between covariance function and sparsity

Complexity Reduction

- Woodbury matrix identity

$$[\boldsymbol{\Sigma}_{\mathrm{M}} + \boldsymbol{\Sigma}_{\varepsilon}]^{-1} = \underbrace{\boldsymbol{\Sigma}_{\mathrm{M}}^{-1}}_{\text{known}} + \underbrace{\boldsymbol{\Sigma}_{\mathrm{M}}^{-1}}_{\text{sparse}} \Big[\underbrace{\boldsymbol{\Sigma}_{\mathrm{M}}^{-1} + \boldsymbol{\Sigma}_{\varepsilon}^{-1}}_{\text{sparse}}\Big]^{-1} \boldsymbol{\Sigma}_{\mathrm{M}}^{-1}$$

  - inversion: $\mathcal{O}(n^2)$
  - multiplications: $\mathcal{O}(n^2)$
  - addition: $\mathcal{O}(n^2)$

- It takes $\mathcal{O}(n^2)$ time to compute BLUP

$$\widehat{\mathrm{Z}}(\boldsymbol{x}_0) = \beta + \underbrace{\boldsymbol{\Sigma}_{\mathrm{M}}(\boldsymbol{x}_0, \cdot)}_{\text{known}}[\boldsymbol{\Sigma}_{\mathrm{M}} + \boldsymbol{\Sigma}_{\varepsilon}]^{-1}[\overline{\boldsymbol{z}} - \beta \mathbf{1}_n]$$

  - If the noise is negligible ($\boldsymbol{\Sigma}_{\varepsilon} \approx \mathbf{0}$), then no numerical inversion is needed and computing BLUP is $\mathcal{O}(n)$!

Stability Improvement

1. $\boldsymbol{\Sigma}_{\mathrm{M}}$ can be made much better conditioned
2. Woodbury also improves numerical stability

$$[\boldsymbol{\Sigma}_{\mathrm{M}} + \boldsymbol{\Sigma}_{\varepsilon}]^{-1} = \boldsymbol{\Sigma}_{\mathrm{M}}^{-1} + \boldsymbol{\Sigma}_{\mathrm{M}}^{-1}\left[\boldsymbol{\Sigma}_{\mathrm{M}}^{-1} + \boldsymbol{\Sigma}_{\varepsilon}^{-1}\right]^{-1}\boldsymbol{\Sigma}_{\mathrm{M}}^{-1}$$

- The diagonal entries of $\boldsymbol{\Sigma}_{\varepsilon}^{-1}$ are often large

1-D Markovian Gaussian Processes

- Assume $\mathcal{X} = [0, 1]$ and $x_i = \frac{i}{n+1}$, $i = 1, \ldots, n$
- Brownian motion: $k(x, y) = \min(x, y)$

$$\boldsymbol{\Sigma}_{\mathrm{M}}^{-1} = (n+1) \begin{pmatrix} 2 & -1 & & & \\ -1 & 2 & -1 & & \\ & \ddots & \ddots & \ddots & \\ & & -1 & 2 & -1 \\ & & & -1 & 1 \end{pmatrix}$$

- Brownian bridge: $k(x, y) = \min(x, y) - xy$

$$\boldsymbol{\Sigma}_{\mathrm{M}}^{-1} = (n+1) \begin{pmatrix} 2 & -1 & & & \\ -1 & 2 & -1 & & \\ & \ddots & \ddots & \ddots & \\ & & -1 & 2 & -1 \\ & & & -1 & 2 \end{pmatrix}$$

- Ornstein-Uhlenbeck process: $k(x,y) = \frac{\sigma^2}{2\theta} e^{-\theta|x-y|}$

$$\mathrm{d}X(t) = -\theta X(t)\,\mathrm{d}t + \sigma \mathrm{d}B(t)$$

$$\boldsymbol{\Sigma}_{\mathrm{M}}^{-1} = \frac{\theta}{\sigma^2 \sinh(\theta h)} \begin{pmatrix} e^{\theta h} & -1 & & & \\ -1 & 2\cosh(\theta h) & -1 & & \\ & \ddots & \ddots & \ddots & \\ & & -1 & 2\cosh(\theta h) & -1 \\ & & & -1 & e^{\theta h} \end{pmatrix},$$

with $h = 1/(n+1)$

Key Observation

- Share the same functional form

$$k(x, y) = p(x)q(y)\,\mathbb{I}_{\{x \le y\}} + p(y)q(x)\,\mathbb{I}_{\{x > y\}},$$

  for some functions f and g
- $k_{\mathrm{BM}}(x, y) = x\,\mathbb{I}_{\{x \le y\}} + y\,\mathbb{I}_{\{x > y\}}$
- $k_{\mathrm{BR}}(x, y) = x(1 - y)\,\mathbb{I}_{\{x \le y\}} + y(1 - x)\,\mathbb{I}_{\{x > y\}}$
- $k_{\mathrm{OU}}(x, y) = e^{x} e^{-y}\,\mathbb{I}_{\{x \le y\}} + e^{y} e^{-x}\,\mathbb{I}_{\{x > y\}}$

Key Observation

- Share the same functional form

$$k(x,y) = p(x)q(y)\,\mathbb{I}_{\{x\le y\}} + p(y)q(x)\,\mathbb{I}_{\{x>y\}},$$

  for some functions f and g
- $k_{\mathrm{BM}}(x,y) = x\,\mathbb{I}_{\{x\le y\}} + y\,\mathbb{I}_{\{x>y\}}$
- $k_{\mathrm{BR}}(x,y) = x(1-y)\,\mathbb{I}_{\{x\le y\}} + y(1-x)\,\mathbb{I}_{\{x>y\}}$
- $k_{\mathrm{OU}}(x,y) = e^{x}e^{-y}\,\mathbb{I}_{\{x\le y\}} + e^{y}e^{-x}\,\mathbb{I}_{\{x>y\}}$

Theorem (Ding and Z, 2017)

If $\boldsymbol{K}$ is nonsingular, then $\boldsymbol{K}^{-1}$ is tridiagonal.

- $\{x_1,\ldots,x_n\}$ are not necessarily equally spaced

Proof by Linear Algebra

- Show $(\boldsymbol{K}^{-1})_{i,j} = 1$ for $|j - i| \geq 2$ by induction on n
- Relation between the inverse and the minors

$$(\boldsymbol{K})^{-1}_{i,j} = \frac{1}{|\boldsymbol{K}|}(-1)^{i+j} M_{j,i}$$

- Use the Laplace expansion of the determinant

$$|\boldsymbol{K}| = \sum_{\ell=1}^{n} (-1)^{i+\ell} k_{i,l} M_{i,l} = \sum_{\ell=1}^{n} (-1)^{\ell+j} k_{\ell,j} M_{\ell,j}$$

Positive Definiteness

- What conditions make $\boldsymbol{K}$ positive definite (PD)?

Positive Definiteness

- What conditions make $\boldsymbol{K}$ positive definite (PD)?

Theorem (Ding and Z, 2017)

1. For $n \geq 2$,
   $|\boldsymbol{K}| = p(x_1)q(x_n)\prod_{i=1}^{n}[p(x_i)q(x_{i-1}) - p(x_{i-1})q(x_i)]$.

Positive Definiteness

- What conditions make $\boldsymbol{K}$ positive definite (PD)?

Theorem (Ding and Z, 2017)

1. For $n \geq 2$,
   $|\boldsymbol{K}| = p(x_1)q(x_n)\prod_{i=1}^{n}[p(x_i)q(x_{i-1}) - p(x_{i-1})q(x_i)]$.
2. $k(x, y)$ is PD *if and only if*
   - $p(x)q(y) - p(y)q(x) < 0$ for all $x < y$;
   - $p(x)q(y) > 0$ for all x, y.

- We call such $k(x, y)$ (1-dimensional) Markovian covariance function (MCF)

A Naïve Example

- Let $q(x) \equiv 1$ and $p(x)$ be positive, strictly increasing
  - $k(x, y) = p(x)\,\mathbb{I}_{\{x \leq y\}} + p(y)\,\mathbb{I}_{\{x > y\}}$, not very reasonable
  - $k(x, y) = \min(p(x), p(y))$: "time-changed" Brownian motion

A Naïve Example

- Let $q(x) \equiv 1$ and $p(x)$ be positive, strictly increasing
  - $k(x, y) = p(x)\,\mathbb{I}_{\{x \le y\}} + p(y)\,\mathbb{I}_{\{x > y\}}$, not very reasonable
  - $k(x, y) = \min(p(x), p(y))$: "time-changed" Brownian motion

- How to construct MCFs easily?

The Green's function of a Sturm-Liouville equation has exactly the same form!

Sturm-Liouville (S-L) Problem

S-L Differential Equation

- Assume $\mathcal{X} = [L, R]$

$$\mathcal{L}f(x) := \frac{1}{w(x)}\left[\frac{\mathrm{d}}{\mathrm{d}x}\left(-u(x)\frac{\mathrm{d}f(x)}{\mathrm{d}x}\right) + v(x)f(x)\right] = h(x),$$

  with boundary condition

$$\begin{cases} \alpha_L f(L) + \beta_L f'(L) = 0, \\ \alpha_R f(R) + \beta_R f'(R) = 0. \end{cases}$$

- Any second-order linear ordinary differential equations can be recast in the form of the S-L equation

Green's Function

- The Green's function satisfies $\mathcal{L}G(x,y) = \delta(x-y)$ with the same BC
- The S-L solution is $f(x) = \int_L^R G(x,y)h(y)\mathrm{d}y$
- Form of Green's function:

$$G(x,y) = cf_1(x)f_2(y)\,\mathbb{I}_{\{x\le y\}} + cf_1(y)f_2(x)\,\mathbb{I}_{\{x>y\}},$$

for some constant c, where $\mathcal{L}f_i = 0$, with

$$\alpha_L f_1(L) + \beta_L f_1'(L) = 0 \quad \text{and} \quad \alpha_R f_2(R) + \beta_R f_2'(R) = 0.$$

Green's Function

- The Green's function satisfies $\mathcal{L}G(x,y) = \delta(x-y)$ with the same BC
- The S-L solution is $f(x) = \int_L^R G(x,y)h(y)\mathrm{d}y$
- Form of Green's function:

$$G(x,y) = cf_1(x)f_2(y)\,\mathbb{I}_{\{x\le y\}} + cf_1(y)f_2(x)\,\mathbb{I}_{\{x>y\}},$$

for some constant c, where $\mathcal{L}f_i = 0$, with

$$\alpha_L f_1(L) + \beta_L f_1'(L) = 0 \quad \text{and} \quad \alpha_R f_2(R) + \beta_R f_2'(R) = 0.$$

- What kind of S-L equations yield a PD Green's function?

The Eigenvalue Problem (S-L Problem)

- $\mathcal{L}f(x) = \lambda f(x)$ with the same boundary condition
- *Regular*: u, u', v, w are continuous, and $u, w > 0$

S-L Theory

The regular S-L problem has a countable number of real eigenvalues and the normalized eigenfunctions can be chosen real-valued and form an orthonormal basis.

- If the eigenvalues are all positive, then
  - $G(x, y) = \sum_\ell \lambda_\ell^{-1} \phi_\ell(x) \phi_\ell(y)$
  - PSD

Positive Eigenvalues

- Integration by parts

$$\lambda_\ell = \langle \mathcal{L}\phi_\ell, \phi_\ell \rangle = - u(x)\phi_\ell(x)\phi_\ell'(x)\Big|_{x=L}^{R} + \int_L^R u(x)[\phi_\ell'(x)]^2 \mathrm{d}x + \int_L^R v(x)[\phi_\ell(x)]^2 \mathrm{d}x$$

- Specify u, v and BC to ensure $\lambda_\ell > 0$
  - E.g., $v > 0$ and the Dirichlet BC $f(L) = f(R) = 0$

Positive Eigenvalues

- Integration by parts

$$\lambda_\ell = \langle \mathcal{L}\phi_\ell, \phi_\ell \rangle = - u(x)\phi_\ell(x)\phi_\ell'(x)\Big|_{x=L}^{R} + \int_L^R u(x)[\phi_\ell'(x)]^2 \mathrm{d}x + \int_L^R v(x)[\phi_\ell(x)]^2 \mathrm{d}x$$

- Specify u, v and BC to ensure $\lambda_\ell > 0$
  - E.g., $v > 0$ and the Dirichlet BC $f(L) = f(R) = 0$

Theorem (Ding and Z, 2017)

Suppose the S-L equation is regular with $v > 0$ and the Dirichlet BC. Then, its Green's function is an MCF.

Examples

- Assume $u(x) \equiv 1$, $v(x) \equiv \nu$, and $w(x) \equiv 1$ on $[0, 1]$:

$$-f''(x) + \nu f(x) = 0$$

	Dirichlet	Cauchy
$\nu = 0$	$p(x) = x$ $q(y) = 1 - y$	$p(x) = x$ $q(y) = 1$
$\nu > 0$	$p(x) = \sinh(\gamma x)$ $q(y) = \sinh(\gamma(1 - y))$	$p(x) = d(\gamma)\sinh(\gamma x)$ $q(y) = \cosh(\gamma(1 - y))$
$\nu < 0$	$p(x) = \sin(\gamma x)$ $q(y) = \sin(\gamma(1 - y))$	$p(x) = d(\gamma)\sin(\gamma x)$ $q(y) = \cos(\gamma(1 - y))$

- $\gamma = \sqrt{|\nu|}$

Corollary (Ding and Z, 2017)

Let $g(x,y) = \eta[p(x)q(y)\,\mathbb{I}_{\{x\le y\}} + p(y)q(x)\,\mathbb{I}_{\{x>y\}}]$ be as given on the last page. Suppose that $\{x_1,\ldots,x_n\}\subset(0,1)$, where $x_i = x_1 + (i-1)h$ with $h = \frac{x_n - x_1}{n-1}$. Then,

$$\boldsymbol{G}^{-1} = \eta^{-1} a \begin{pmatrix} b & -1 & & & \\ -1 & c & -1 & & \\ \cdots & & \cdots & & \cdots \\ & & -1 & c & -1 \\ & & & -1 & d \end{pmatrix}.$$

Corollary (Ding and Z, 2017)

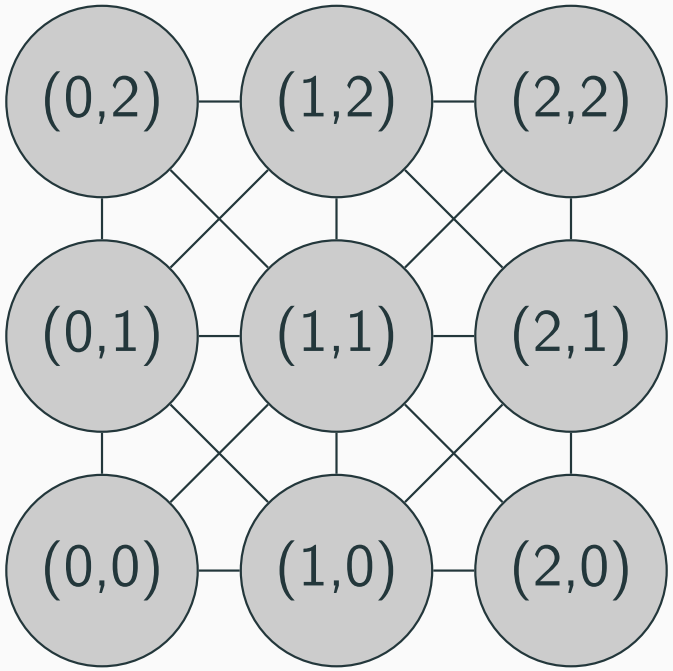
Let $g(x,y) = \eta[p(x)q(y)\,\mathbb{I}_{\{x\le y\}} + p(y)q(x)\,\mathbb{I}_{\{x>y\}}]$ be as given on the last page. Suppose that $\{x_1,\ldots,x_n\}\subset(0,1)$, where $x_i = x_1 + (i-1)h$ with $h = \frac{x_n - x_1}{n-1}$. Then,

$$\boldsymbol{G}^{-1} = \eta^{-1} a \begin{pmatrix} b & -1 & & & \\ -1 & c & -1 & & \\ \cdots & & \cdots & & \cdots \\ & & -1 & c & -1 \\ & & & -1 & d \end{pmatrix}.$$

- $\mathcal{O}(1)$ computation
- Reparameterization

Extension for $d > 1$

- "Compositional" covariance: $k(\boldsymbol{x}, \boldsymbol{y}) = \prod_{i=1}^{d} k_i(x^i, y^i)$
- Limitation: $\{\boldsymbol{x}_1, \ldots, \boldsymbol{x}_n\}$ must form a regular lattice
- Then, $\boldsymbol{K} = \bigotimes_{i=1}^{d} \boldsymbol{K}_i$ and $\boldsymbol{K}^{-1} = \bigotimes_{i=1}^{d} \boldsymbol{K}_i^{-1}$, preserving sparsity



GP with Derivative Information

Partial Derivative of Gaussian Process

- Given a GRF $Z(\mathbf{x})$ with kernel $k(\cdot, \cdot)$, we have

$$\lim_{||h|| \to 0} E|Z(\mathbf{x} + h) - Z(\mathbf{x})|^2 = 0$$

- Its partial derivative is defined in L^2 sense:

$$\frac{dZ(\mathbf{x})}{dx_i} := \lim_{\epsilon \to 0} \frac{Z(\mathbf{x} + \epsilon e_i) - Z(\mathbf{x})}{\epsilon}$$

- Since linear combination of Gaussian random variables is Gaussian and hence the limit on the RHS is Gaussian

- Partial derivative of a GRF is a GRF with covariance kernel:

$$Cov(\frac{dZ(\mathbf{x})}{dx_i}, \frac{dZ(\mathbf{y})}{dy_j}) = \frac{\partial^2 Cov(Z(\mathbf{x}), Z(\mathbf{y}))}{\partial x_i \partial y_j} = \frac{\partial^2 k(\mathbf{x}, \mathbf{y})}{\partial x_i \partial y_j}$$

  given $k(\mathbf{x}, \mathbf{y})$ is differentiable at $\mathbf{x}$ and $\mathbf{y}$

- The covariance between $Z(\mathbf{x})$ and $\frac{dZ(\mathbf{y})}{dy_i}$ is:

$$Cov(Z(\mathbf{x}), \frac{dZ(\mathbf{y})}{dy_i}) = \frac{\partial Cov(Z(\mathbf{x}), Z(\mathbf{y}))}{\partial y_i} = \frac{\partial k(\mathbf{x}, \mathbf{y})}{\partial y_i}$$

Prediction Using Gradient Information

- Given observations of the original GP $(\mathbf{X}, \mathbf{Y})$ and observations of partial derivatives $\{(\mathbf{X}, D^i\mathbf{Y}), i = 1, \cdots, d\}$
- We can write down the covariance matrix:

$$\Sigma_+ = \begin{pmatrix} Cov(\mathbf{Y}, \mathbf{Y}) & Cov(\mathbf{Y}, D^1\mathbf{Y}) & \cdots & Cov(\mathbf{Y}, D^d\mathbf{Y}) \\ Cov(D^1\mathbf{Y}, \mathbf{Y}) & Cov(D^1\mathbf{Y}, D^1\mathbf{Y}) & \cdots & Cov(D^1\mathbf{Y}, D^d\mathbf{Y}) \\ \vdots & & & \\ Cov(D^d\mathbf{Y}, \mathbf{Y}) & Cov(D^d\mathbf{Y}, D^1\mathbf{Y}) & \cdots & Cov(D^d\mathbf{Y}, D^d\mathbf{Y}) \end{pmatrix}$$

$$= \begin{pmatrix} k(\mathbf{X}, \mathbf{X}) & \frac{\partial k(\mathbf{X},\mathbf{X})}{\partial y_1} & \cdots & \frac{\partial k(\mathbf{X},\mathbf{X})}{\partial y_d} \\ \frac{\partial k(\mathbf{X},\mathbf{X})}{\partial x_1} & \frac{\partial^2 k(\mathbf{X},\mathbf{X})}{\partial x_1 \partial y_1} & \cdots & \frac{\partial^2 k(\mathbf{X},\mathbf{X})}{\partial x_1 \partial y_d} \\ \vdots & & & \\ \frac{\partial k(\mathbf{X},\mathbf{X})}{\partial x_d} & \frac{\partial^2 k(\mathbf{X},\mathbf{X})}{\partial x_d \partial y_1} & \cdots & \frac{\partial^2 k(\mathbf{X},\mathbf{X})}{\partial x_d \partial y_d} \end{pmatrix}$$

- $\Sigma_+(x_0, \cdot)$ becomes:

$$\Sigma_+(x_0, \cdot) = \begin{pmatrix} Cov(Z(x_0), \mathbf{Y}) \\ Cov(Z(x_0), D^1\mathbf{Y}) \\ \vdots \\ Cov(Z(x_0), D^d\mathbf{Y}) \end{pmatrix} = \begin{pmatrix} k(x_0, \mathbf{X}) \\ \frac{k(x_0,\mathbf{X})}{\partial y_1} \\ \vdots \\ \frac{\partial k(x_0,\mathbf{X})}{\partial y_d} \end{pmatrix}$$

- The BULP of $Z(x_0)$ is

$$\hat{Z}(x_0) = \begin{pmatrix} \bar{Z}(x_0) \\ \frac{\partial \bar{Z}(x_0)}{\partial x_1} \\ \vdots \\ \frac{\partial \bar{Z}(x_0)}{\partial x_d} \end{pmatrix} + \Sigma_+(x_0, \cdot)\left[\Sigma_+ + \Sigma_\epsilon\right]^{-1} \begin{pmatrix} \mathbf{Y} - \bar{Z}(\mathbf{X}) \\ D^1\mathbf{Y} - \frac{\partial \bar{Z}(\mathbf{X})}{\partial x_1} \\ \vdots \\ D^n\mathbf{Y} - \frac{\partial \bar{Z}(\mathbf{X})}{\partial x_d} \end{pmatrix}$$

- The BULP of $D^i Z(x_0)$ can be computed in a similar way

Issues in Applications

- We meet the Big n Problem again so Stochastic Kriging with Gradient(SKG) is not practical for large data sets unless Σ_+ is sparse
- Markovian kernels give sparsity but we notice that Markovian kernels are not differentiable in a normal sense
- How to construct SKG with sparse kernel matrix?

Weak Derivative of GP

An Elementary Example: Brownian Motion

- We use Brownian motion(BM) as an example: derivative of a BM $\dot{B}(t)$ is defined as the white noise process $\eta(t)$ whose covariance function is given by the Dirac-delta function:

$$\delta(x-y) = \begin{cases} \infty, & \text{if } \ x = y \\ 0, & \text{otherwise} \end{cases} \qquad \int_{-\epsilon}^{\epsilon} \delta(s)ds = 1$$

  for any $\epsilon > 0$

- From mathematical analysis, we get

$$Cov(B(x), \eta(y)) = \mathbb{I}_{\{x>y\}} = \int_0^x \delta(u-y)du$$

$$Cov(B(x), B(y)) = \min(x,y) = \int_0^x \int_0^y \delta(u-v)dudv$$

- $\mathbb{I}_{\{x>y\}}$ and $\delta(x-y)$ are the **weak derivative** of $\min(x,y)$ and $\mathbb{I}_{\{x>y\}}$ respectively

- Suppose we have a discretized BM with observations $(\mathbf{X}, \mathbf{Y})$ and derivative observations $(\mathbf{X}, D\mathbf{Y})$, we can write down the covariance matrix Σ_+:

$$\Sigma_+ = \begin{pmatrix} Cov(\mathbf{Y}, \mathbf{Y}) & Cov(\mathbf{Y}, D\mathbf{Y}) \\ Cov(D\mathbf{Y}, \mathbf{Y}) & Cov(D\mathbf{Y}, D\mathbf{Y}) \end{pmatrix}$$
$$= \begin{pmatrix} \min(\mathbf{X}, \mathbf{X}) & \mathbb{I}_{\{\mathbf{X} > \mathbf{X}\}} \\ \mathbb{I}_{\{\mathbf{X} < \mathbf{X}\}} & Diag([\frac{1}{h_1}, \cdots, \frac{1}{h_N}]) \end{pmatrix}$$

  where $\{h_i\}$ are close to 0 and treated as tuning parameters
- Σ_+^{-1} is a sparse matrix whose non-zero entries can be calculated in closed form

- The essential part in computing Σ_+^{-1} is the Schur complement:

$$S := Diag([\frac{1}{h_1}, \cdots, \frac{1}{h_N}]) - \mathbb{I}_{\{\mathbf{x}<\mathbf{x}\}} \left[\min(\mathbf{X}, \mathbf{X})\right]^{-1} \mathbb{I}_{\{\mathbf{x}>\mathbf{x}\}}$$

  which is a diagonal matrix

- Let

$$\Sigma_+^{-1} := \begin{pmatrix} T_{11} & T_{12} \\ T_{21} & T_{22} \end{pmatrix}$$

  the matrix block inversion identity gives:

$$\begin{aligned} T_{22} &= S^{-1} \\ T_{12} &= T_{21}' = -\left[\min(\mathbf{X}, \mathbf{X})\right]^{-1} \mathbb{I}_{\{\mathbf{x}>\mathbf{x}\}} S^{-1} \\ T_{11} &= \left[\min(\mathbf{X}, \mathbf{X})\right]^{-1} \\ &\quad + \left[\min(\mathbf{X}, \mathbf{X})\right]^{-1} \mathbb{I}_{\{\mathbf{x}>\mathbf{x}\}} S^{-1} \mathbb{I}_{\{\mathbf{x}<\mathbf{x}\}} \left[\min(\mathbf{X}, \mathbf{X})\right]^{-1} \end{aligned}$$

- T_{11} is tridiagonal; T_{22} is diagonal

Another Example: the Langevin Equation

- We construct a more general model based on stochastic differential equation(SDE)
- The Langevin equation:

$$\dot{V}(t) = -\theta V(t) + \epsilon\eta(t)$$

  ($V(t)$ becomes a BM when $\theta = 0$)
- Rather than treat $V(t)$ as a stochastic process, we treat $V(t)$ as a 1-D Gaussian random field:

  What are the covariance functions of $V(t)$ and $\dot{V}(t)$?

- Averaging the equation:

$$\mu(t) := E[V(t)] = -\theta E[\dot{V}(t)] = -\theta \dot{\mu}(t)$$

$$\Longrightarrow$$

$$\mu(t) = \mu(t_0) e^{-\theta(t-t_0)}$$

- Define the covariance functions:

$$g(x,y) := E[\{V(x) - \mu(x)\}\{V(y) - \mu(y)\}]$$
$$\frac{\partial^2 g}{\partial x \partial y}(x,y) := E[\{\dot{V}(x) - \dot{\mu}(x)\}\{\dot{V}(y) - \dot{\mu}(y)\}]$$
$$\frac{\partial g}{\partial y}(x,y) := E[\{V(x) - \mu(x)\}\{\dot{V}(y) - \dot{\mu}(y)\}]$$

- Plug Langevin in the above equations, we have the PDE:

$$\frac{\partial^2 g}{\partial x \partial y} = -\theta^2 g - \theta(\frac{\partial g}{\partial x} + \frac{\partial g}{\partial y}) + \epsilon \delta(x-y)$$

where the derivative is defined in the weak sense

- By imposing different boundary conditions on the PDE, we can solve for different kinds of Markovian kernel. For example, the OU kernel:

$$
\begin{aligned}
g(x,y) &= \frac{\epsilon^2}{2\theta} e^{-\theta|x-y|} \\
\frac{\partial^2 g}{\partial x \partial y}(x,y) &= \frac{\epsilon^2}{2\theta}\left[2\theta\delta(x-y) - \theta^2 e^{-\theta|x-y|}\right] \\
\frac{\partial g}{\partial y}(x,y) &= \epsilon^2 e^{-\theta(x-y)}\, \mathbb{I}_{\{x>y\}} - \theta g(x,y)
\end{aligned}
$$

- We run through the same matrix algebra and it turns out the Σ_+^{-1} of OU shares the same sparsity:

$$
\Sigma_+^{-1} = \begin{pmatrix} g(\mathbf{X},\mathbf{X}) & \frac{\partial g}{\partial y}(\mathbf{X},\mathbf{X}) \\ \frac{\partial g}{\partial x}(\mathbf{X},\mathbf{X}) & \frac{\partial^2 g}{\partial x \partial y}(\mathbf{X},\mathbf{X}) \end{pmatrix}^{-1} =: \begin{pmatrix} T_{11} & T_{12} \\ T_{21} & T_{22} \end{pmatrix}
$$

where T_{11} is tridiagonal and T_{22} is diagonal

Extension for d>1

- 'Compositional' kernel: $g(\mathbf{x}, \mathbf{y}) = \prod_{i=1}^{d} g_i(x_i, y_i)$
- the partial derivative of $g(x, y)$ is still compositional:

$$\frac{\partial^2 g}{\partial x_s \partial y_t}(x, y) = \frac{\partial g_s}{\partial x_s}(x_s, y_s) \frac{\partial g_t}{\partial y_t}(x_t, y_t) \prod_{i \neq s,t}^{d} g_i(x_i, y_i)$$

- If $\{\mathbf{x}_1, \cdots, \mathbf{x}_n\}$ form a lattice, we have:

$$\Sigma_+ = \begin{pmatrix} C_{0,0} & C_{0,d} & C_{0,d-1} & \cdots & C_{0,1} \\ C_{d,0} & C_{d,d} & C_{d,d-1} & \cdots & C_{d,1} \\ \vdots & & & & \\ C_{1,0} & C_{1,d} & \cdots & & C_{1,1} \end{pmatrix}$$

where

$$C_{0,0} = \bigotimes_{i=1}^{d} G_i,$$

$$C_{0,k} = \{\bigotimes_{i=1}^{k-1} G_i\} \bigotimes \{\frac{\partial G_k}{\partial y_k}\} \bigotimes \{\bigotimes_{i=k+1}^{d} G_i\}$$

$$C_{k,0} = C_{0,k}^T = \{\bigotimes_{i=1}^{k-1} G_i\} \bigotimes \{\frac{\partial G_k}{\partial x_k}\} \bigotimes \{\bigotimes_{i=k+1}^{d} G_i\}$$

$$C_{k,k} = \{\bigotimes_{i=1}^{k-1} G_i\} \bigotimes \{\frac{\partial^2 G_k}{\partial x_k \partial y_k}\} \bigotimes \{\bigotimes_{i=k+1}^{d} G_i\}$$

$$C_{k,l} = \{\bigotimes_{i=1}^{k-1} G_i\} \bigotimes \{\frac{\partial G_k}{\partial x_k}\} \bigotimes \{\bigotimes_{i=k+1}^{l-1} G_i\} \bigotimes \{\frac{\partial G_l}{\partial y_l}\} \bigotimes \{\bigotimes_{i=l+1}^{d} G_i\}$$

- Σ_+ looks messy but, in fact, its inverse is sparse and can be computed in linear time complexity
- Let's define

$$T^{(k)} = \begin{pmatrix} T_{11}^{(k)} & T_{12}^{(k)} \\ T_{21}^{(k)} & T_{22}^{(k)} \end{pmatrix} := \begin{pmatrix} G_k & \frac{\partial G_k}{\partial y_k} \\ \frac{\partial G_k}{\partial x_k} & \frac{\partial^2 G_k}{\partial x_k \partial y_k} \end{pmatrix}^{-1}$$

and define

$$\Sigma_+^{-1} = T = \begin{pmatrix} T_{11} & T_{12} \\ T_{21} & T_{22} \end{pmatrix}$$

then

$$T_{11} = (1-d) \bigotimes_{i=1}^{d} G_i^{-1} + \sum_{i=1}^{d} \{ \bigotimes_{j=1}^{i-1} G_i^{-1} \} \bigotimes T_{11}^{(i)} \bigotimes \{ \bigotimes_{j=i+1}^{d} G_i^{-1} \}$$

$$T_{22} = diag([\{ \bigotimes_{j=1}^{i-1} G_i^{-1} \} \bigotimes T_{22}^{(i)} \bigotimes \{ \bigotimes_{j=i+1}^{d} G_i^{-1} \}]_{i=d}^{1})$$

$$T_{12} = T_{21}' = [\{ \bigotimes_{j=1}^{i-1} G_i^{-1} \} \bigotimes T_{12}^{(i)} \bigotimes \{ \bigotimes_{j=i+1}^{d} G_i^{-1} \}]_{i=d}^{1}$$

- From previous content, we know G_k^{-1} and $T_{ij}^{(k)}$ are sparse and can be computed explicitly so does T
- In dimension two:

$$
\begin{aligned}
T_{11} &= G_1^{-1} \bigotimes T_{11}^{(2)} + T_{11}^{(1)} \bigotimes G_2^{-1} - G_1^{-1} \bigotimes G_2^{-1} \\
T_{12} &= \begin{bmatrix} G_1^{-1} \bigotimes T_{12}^{(2)} & T_{12}^{(1)} \bigotimes G_2^{-1} \end{bmatrix} \\
T_{22} &= \begin{pmatrix} G_1^{-1} \bigotimes T_{22}^{(2)} & 0 \\ 0 & T_{22}^{(1)} \bigotimes G_2^{-1} \end{pmatrix}
\end{aligned}
$$

Remarks

- We have checked the Σ_+^{-1} of some Markov kernels and all of them share the same sparsity but we dot not know if this holds for **ALL** 1-D Markov kernels
- We look into the matrix Σ_+^{-1} and believe that the answer is Yes
- Even if the answer is no, Σ_+^{-1} in dimension two or higher is still sparse if all design points form a lattice
- Ito's Lemma and weak derivative are equivalent: when we take the derivative of a SDE according to Ito's Lemma, it turns out the covariance function of the former is the weak derivative of the later

Numerical Experiments

Three Covariance Functions

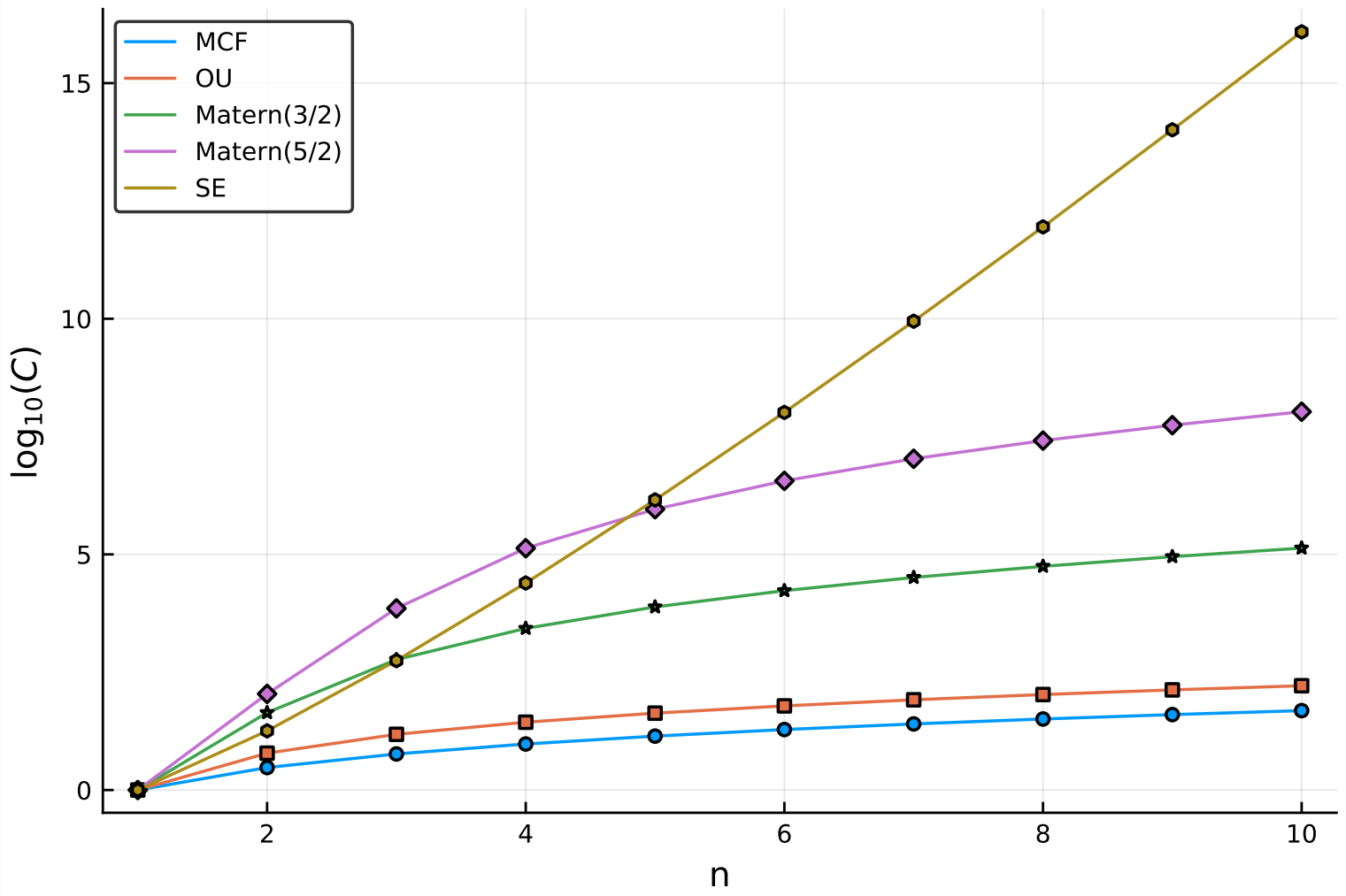
- MCF: $k(x,y) = \eta[p(x)q(y)\,\mathbb{I}_{\{x \le y\}} + p(y)q(x)\,\mathbb{I}_{\{x > y\}}]$

$$\begin{cases} p(x) = \sin(\sqrt{|\nu|}x), & q(x) = \sin(\sqrt{|\nu|}(1-x)), & \text{if } \nu < 0 \\ p(x) = x, & q(x) = 1 - x, & \text{if } \nu = 0 \\ p(x) = \sinh(\sqrt{\nu}x), & q(x) = \sinh(\sqrt{\nu}(1-x)), & \text{if } \nu > 0 \end{cases}$$

- OU: $k(x,y) = \eta \exp(-\theta|x-y|)$
- SE (squared exponential): $k(x,y) = \eta \exp(-\theta(x-y)^2)$

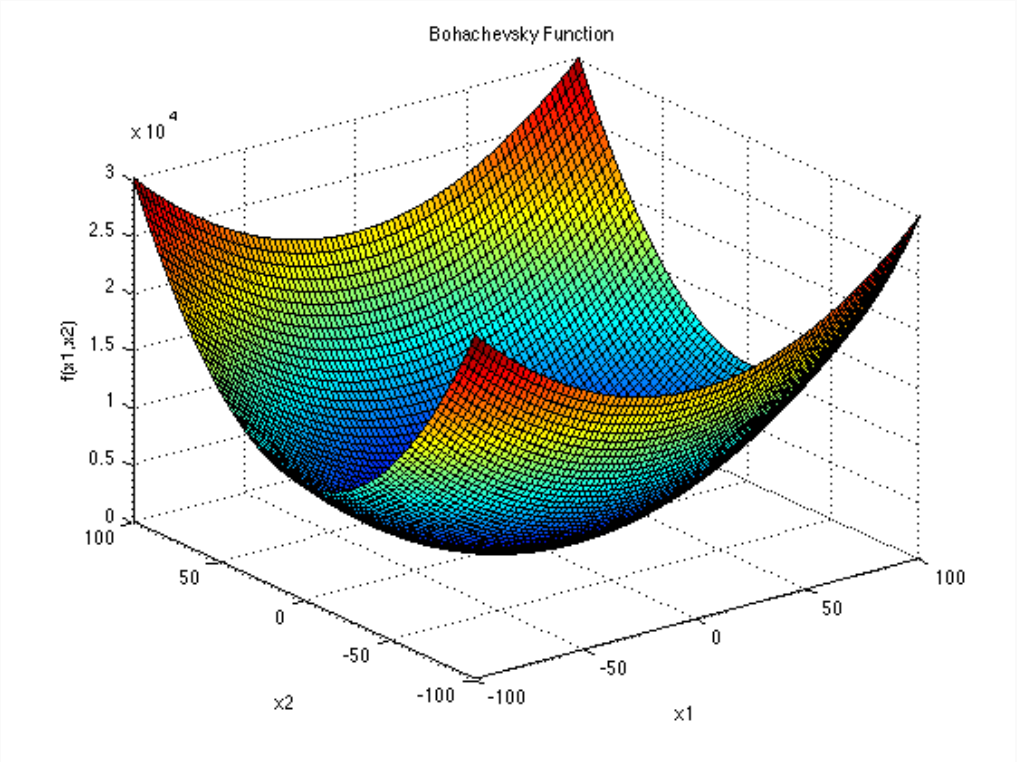
Condition Number of Covariance Matrix

- $C = \lambda_{\max}(\boldsymbol{K})/\lambda_{\min}(\boldsymbol{K})$ measures "closeness to singularity"

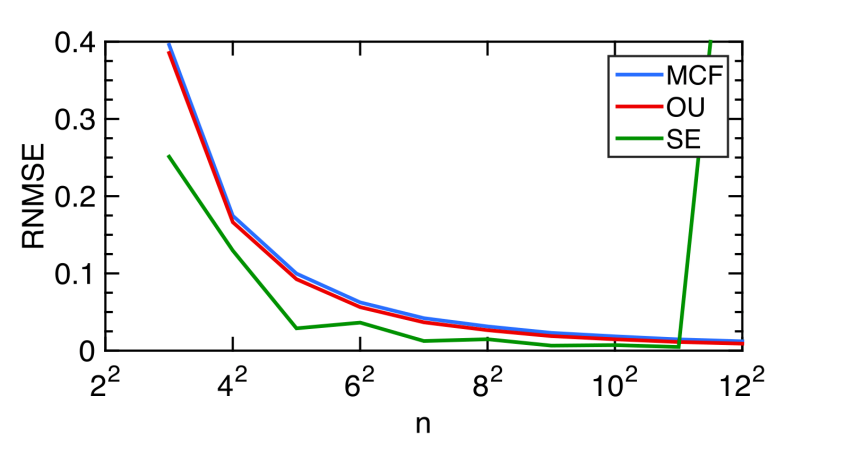


Artificial Surface: Bohachevsky Function

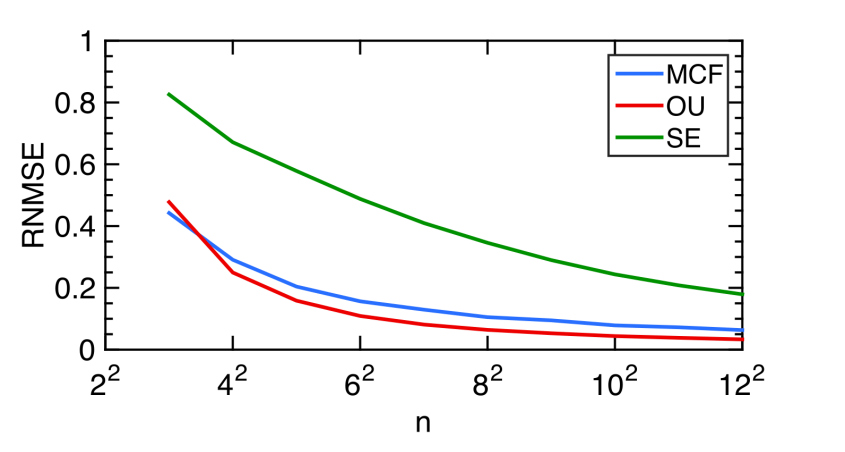
$$f(x, y) = x^2 + 2y^2 - 0.3\cos(3\pi x) - 0.4\cos(4\pi y) + 0.7$$



Prediction



Noise-free Samples



Noisy Samples

Jackson Network

- An N-station Jackson network
- The arrivals consist of D different types of products and the fraction of product i is α_i, $i = 1, \cdots, D$.
- Station j has a service rate μ_j , regardless of the product type, $j = 1, \cdots, N$.
- Let ρ denote the utilization of the bottleneck station.
- The design variable is $(\alpha_1, \cdots, \alpha_D)$, for $\alpha_i \in [0, 1]$ with $\sum_{i=1}^{D} \alpha_i = 1$ and $\rho \in [0.5, 0.9]$.
- The value of interest is the cycle time(CT) defined given by:

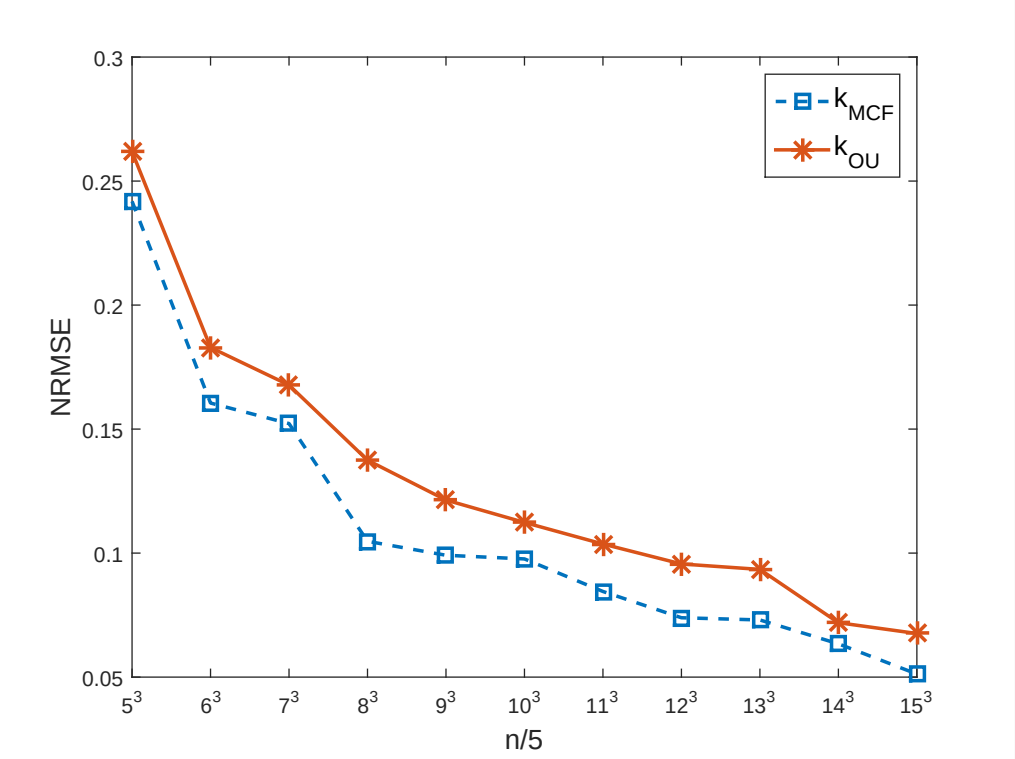
$$\mathbb{E}[CT_k] = \sum_{j=1}^{N} \frac{\delta_{kj}}{\mu_j \left[1 - \rho\left(\frac{\sum_{i=1}^{D} \alpha_i \delta_{ij}/\mu_j}{\max_h \sum_{i=1}^{D} \alpha_i \delta_{ih}/\mu_h}\right)\right]}$$

- The design variables $\{\alpha_i\}$ cannot be put on a lattice. To fix the issue, we only need a transformation:

$$x_1 = \sqrt{\alpha_1}$$

$$x_i = \sqrt{\frac{\alpha_i}{1 - \sum_{j=1}^{i-1} \alpha_j}}, \quad i = 2, \cdots, D-1$$

so the variable $\mathbf{x} \in [0,1]^{D-1}$



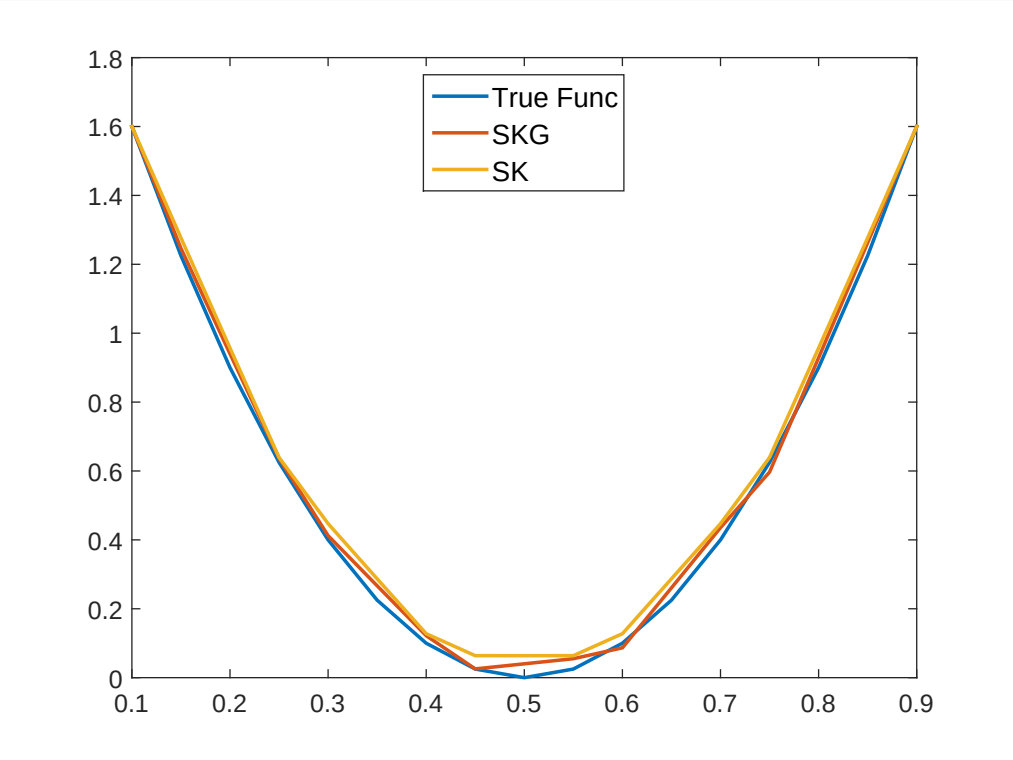
SKG with Gradient Information

- OU kernel is used for comparison between stochastic Kriging with gradient(SKG) and stochastic kriging(SK):

$$
\begin{aligned}
g(x,y) &= \frac{\epsilon^2}{2\theta} e^{-\theta|x-y|} \\
\frac{\partial^2 g}{\partial x \partial y}(x,y) &= \frac{\epsilon^2}{2\theta}\left[2\theta\delta(x-y) - \theta^2 e^{-\theta|x-y|}\right] \\
\frac{\partial g}{\partial y}(x,y) &= \epsilon^2 e^{-\theta(x-y)}\, \mathbb{I}_{\{x>y\}} - \theta g(x,y)
\end{aligned}
$$

Artificial 1-D Function

$$f(x) = 10(x - 0.5)^2$$
$$f'(x) = 20(x - 0.5)$$

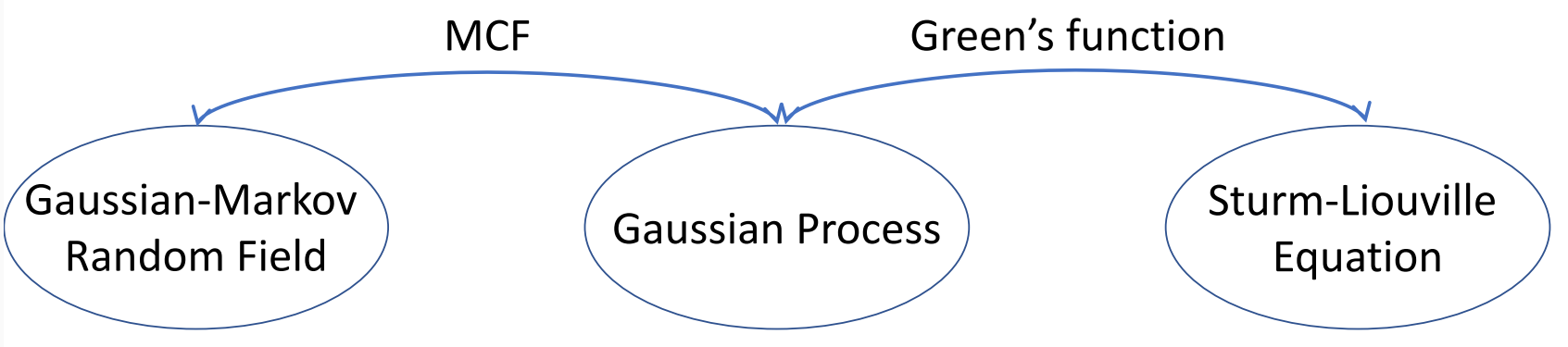


Tests in Higher Dimension

Test	Dim	SK data #	SKG data #	Improvement
$\exp(-\|\|x\|\|^2)$	two	10^2	3×10^2	96%
$\exp(-\|\|x\|\|^2)$	six	4^6	7×4^6	41%
$\prod_i^D x_i$	six	4^6	7×4^6	38%
Closed Loop Assembly System	six	4^6	7×4^6	29%

Conclusions

Conclusions



- MCFs allow modeling association directly, while retaining sparsity in the precision matrix
- Reduce computational cost from $\mathcal{O}(n^3)$ to $\mathcal{O}(n^2)$ without approximations
  - Further reduce to $\mathcal{O}(n)$ if observations are noise-free
- Enhance numerical stability substantially
- Limitation: design points must form a regular lattice, though not necessarily equally spaced

Markovian covariances without approx.

v.s.

Good approx. for *all* covariances